

Roll No.

TME-501

**B. TECH. (ME) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

HEAT AND MASS TRANSFER

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the following in brief : 10 Marks (CO1)
- (i) Critical thickness of insulation
- (ii) Effect of temperature on thermal conductivity of solid, liquid and gases 10 Marks (CO1)

OR

- (b) Define and explain steady, unsteady, uniform and non-uniform heat transfer with suitable equation. 10 Marks (CO1)
2. (a) Derive the critical radius of insulation for sphere. 10 Marks (CO1)

OR

- (b) Differentiate between Biot number and Nusselt number with relevant example. 10 Marks (CO1)

P. T. O.

3. (a) Determine the heat transfer rate across a composite slab which is made of different materials with top and bottom. The entire left-hand face is held at the temperature T_1 while the entire right hand face is at the temperature T_2 . The conductivities of the two different materials are stated as k_a and k_b and their areas as viewed in the direction of slab thickness δ are A_a and A_b respectively. Steady state exists, there is no heat generation and the slab is so thin that any edge effects can be neglected. Interpret the result in terms of an electrical circuit.

10 Marks (CO1/CO2)

OR

- (b) The hot combustion gases at 300°C flow through a hollow cylindrical pipe of 10 cm inner diameter and 12 cm outer diameter. The pipe is located in a space at 30°C and the thermal conductivity of the pipe material is 200 W/mK . Neglecting surface heat transfer coefficients, calculate the heat loss through the pipe per unit length and the temperature at a point halfway between the inner and outer surface. What should be the surface area normal to the direction of heat flow so that the heat transfer through the pipe can be determined by considering material of the pipe as a plane wall of the same thickness ?

10 Marks (CO1/CO2)

4. (a) Two insulation materials A and B, in powder form, with thermal conductivities of 0.005 W/m-deg and 0.03 W/m-deg were purchased for use over a sphere of 50 cm diameter. Material A was to form the first layer 4 cm thick and material B was to be the next layer 5 cm thick. Due to oversight during installation, whole of material B was applied first and subsequently there was a layer formed by material A. Investigate how the conduction heat transfer would be affected.

10 Marks (CO1/CO2)

(3)

OR

- (b) A 6.5 m diameter vertical kiln has a hemi-spherical dome at the top; the dome is fabricated from a 25 cm thick layer of chrome brick which has a thermal conductivity of 1.16 W/m-deg. The kiln dome has inside temperature of 875°C and 10°C atmospheric air result into 11.4 W/m²-deg heat transfer coefficient between the dome and air. Estimate the outside surface temperature of the dome and the heat loss from kiln. Compare this heat with that would result from a flat dome fabricated from the same material and with kiln operating under identical temperature conditions. 10 Marks (CO1/CO2)
5. (a) What is lumped heat analysis ? Derive the equation for heat transfer in lumped heat analysis. 10 Marks (CO2)

OR

- (b) A sphere of radius 10 mm at a temperature of 600 K is dipped into the liquid at 300 Kelvin with a convective coefficient of 100 W/mK. Find the time required to become temperature of the sphere equals to 500 K. The sphere has : $C = 375 \text{ J/kg-K}$ 10 Marks (CO2)
- $\rho = 8700 \text{ kg/m}^3$
- $k = 360 \text{ W/mK}$.